

Vishwa Patel

(213)-829-3399 | vishwasa@usc.edu | [linkedin.com/in/vishwapatel2405](https://www.linkedin.com/in/vishwapatel2405) | github.com/vishwapatel24051

Education

University of Southern California, CA, USA

January 2026–December 2027

- Master of Science (M.S.), Computer Science

GPA: 3.66/4

- Courses: Analysis of Algorithms, Web Technologies, Database Systems, Agentic AI, Natural Language Processing

Gujarat Technological University, Gujarat, India

August 2019–June 2023

- Bachelor of Engineering (B.E.), Computer Engineering

GPA: 8.98/10

- Courses: Data Structures, Object-Oriented Programming Principles, Theory of Computation, Compiler Design, Software Engineering, Machine Learning, Artificial Intelligence, Operating Systems

Technical Skills

- **Programming Languages:** Java, C++, Python, C, C#, JavaScript, HTML, CSS, Dart
- **Databases:** MySQL, SQLite, PostgreSQL (incl. pgvector), HBase, Cassandra
- **Frameworks, Libraries & Tools:** PyTorch, TensorFlow, ReactJS, FastAPI, Spring Boot, Spring MVC, Redis, Elasticsearch, RabbitMQ, Docker, Git, jQuery, Flutter

Professional Experience

Casepoint Pvt. Ltd, Gujarat, India

Software Engineer

May 2025–June 2025

- Added custom field selection to the legal hold reporting module so clients could build their own reports, cutting report generation time by 30% while staying within regulatory compliance
- Handled 10–20 high-priority support tickets a week for file extraction and processing failures, digging into root causes and shipping code-level fixes to stay within client SLAs
- Built a companion agent that tracks audio-peak collection progress in real time by request ID, giving users visibility into status and expected completion time

Associate Software Engineer

July 2023–May 2025

- Rebuilt the extraction workflow to run in parallel instead of single-threaded, boosting throughput by ~90%
- Developed a PST repair service with automated recovery and retry logic for corrupted mailbox files, cutting manual intervention by 50–70% and plugging directly into the existing extraction pipeline
- Used Redis to track document status across 6+ processing stages – each record is staged in Redis during bulk ingestion and only written to the database once extraction finishes, which brought status-update latency down by 80%
- Extended the metadata extraction service to support a new file format for forensic analysis and folded it into the existing pipeline
- Upgraded core extraction libraries (Oracle, Aspose, VintaSoft) for better format compatibility, which cut extraction errors and failures by 40%+
- Worked with 5+ teams across product and engineering to scope requirements and ship 8+ production features, including exposing our extraction data to other teams' dashboards without breaking their integrations

Software Engineer Intern

January 2023–June 2023

- Created a desktop-based Recruitment System (.NET MVC, C#, Windows Forms, PostgreSQL, Kendo UI) with 20+ UI screens and both client- and server-side validation for better maintainability
- Added RabbitMQ for asynchronous messaging between app components as part of an Agile sprint cycle

Amiseq, Pune, India

Software Engineering Intern (Backend)

June 2022–July 2022

- Shipped 10+ RESTful API endpoints in Spring Boot and PostgreSQL, structured as modular microservices for CRUD operations
- Tested and debugged those endpoints in Postman, catching integration issues early and validating request/response handling

Projects

GTM Signal Agent | Python, FastAPI, React, PostgreSQL, Groq/Ollama/Anthropic | [GitHub](#)

- Built an AI sales agent (FastAPI, PostgreSQL) that turns company funding, hiring, and tech-stack signals into outreach briefs, with a model layer that works across 3 backends – Groq, Ollama, and Anthropic
- Added a risk-scoring engine that flags deals going stale, so reps know within a rolling 30-day window which accounts need follow-up

Smart Document Reader | Python, FastAPI, PostgreSQL/pgvector, Ollama | [GitHub](#)

- Designed a local-first RAG assistant that combines format-aware extraction with hybrid dense/lexical retrieval to parse PDFs and tables accurately, grounding every answer with a citation back to its exact source
- Tested retrieval accuracy across 16 configurations to land on the best-performing pipeline

TechHire: AI Job Search Platform | Python, FastAPI, React, PostgreSQL | [GitHub](#)

- Built a full-stack job platform (200+ live postings) with filtering by skills, salary, visa sponsorship, and work mode – the search API averages ~7ms across the full index
- Paired it with an AI resume analyzer (Groq/Llama, 3-model pipeline) that scores candidate fit 0–100 and rewrites weak bullets; in blind testing it showed an 85-point spread between strong- and weak-fit resumes

Sequence Alignment | Python | [GitHub](#)

- Implemented global DNA sequence alignment with Needleman-Wunsch DP and Hirschberg's divide-and-conquer, cutting space complexity from $O(mn)$ to $O(m+n)$ with no change in the optimal alignment output
- Measured CPU time and peak memory usage across different sequence lengths and checked correctness against a brute-force baseline

Certifications & Awards

- Python for Everybody Specialization (Coursera) — Certification in Python, databases, web scraping, and data visualization
- Casepoint Excellence Award 2024 (Sprint Delivery Standout) — Awarded for outstanding sprint delivery and performance